

11.429/15.022
Take home final
Due 3/22/2026 5pm

I. Goal: This assignment asks you to compare 8 office markets in the US. The goal is to pick the best and worst markets in terms of how current pricing (as of 2017:3) compares to “fundamental” asset prices, when the latter results from a statistical model using a straightforward Gordon Growth Formula (GGF).

II. Data: the File Cap_panel.xlsx contains data for 8 Metropolitan areas both on historic pricing (NCREIF cap rates) together with historic market fundamentals quarterly, from 1987:1 through 2017:3. Variables are:

- comp: new completions of office space (1000s sqft) (COMP).
- Stock of existing office space (1000s of square feet) (STK)
- average rent/sqft signed in that period (R)
- US cpi (for inflation adjustments) (CPI)
- US 10-year treasury rate (T10)

III. Research Plan/Methodology.

- Separately for each market estimate an Error Correction Model time series regression explaining that period’s cap rate on its value 1 year (4 quarters) prior and: the 10-year interest rate (T10) and the growth in property rental income (R) over that year. A 4 period ECM adjustment is used because of seasonality in NCREIF data. Note also that this model implicitly incorporates a measure of overall historic market risk in its intercept but does not capture any time varying measure of market risk.

$$CAP_t = \beta + \alpha_1 CAP_{t-4} + \alpha_2 T10_t + \alpha_3 (R_t/R_{t-4})$$

- In the strict GGM formulation, the coefficient of the 10-year would be similar or even identical (but of opposite sign) to that on rent growth. Why do you think that is not true here?
- Now expand the model to add in the CPI inflation rate over the last year in addition to property rent growth.

$$CAP_t = \beta + \alpha_1 CAP_{t-4} + \alpha_2 T10_t + \alpha_3 (R_t/R_{t-4}) + \alpha_4 (CPI_t/CPI_{t-4})$$

- In this version the coefficient on rent growth is effectively that for *real* rent growth, since CPI growth is controlled for. Why do you think these two coefficients are so different in magnitude (although both are generally significant)?
- Is estimated *long run* coefficient on 10-year near to 1 as in the GGF?

- How similar now is the coefficient on the 10-year to the average of the coefficient on rent growth and that on inflation.

IV. Investment Criteria

- Regression residuals (actual – predicted) must sum to zero over the full sample by construction.
- Examine the residuals over the last 2 years of the sample as an indication of how “current” market pricing compares to the GGM prediction – estimated specifically from that market’s historic data.
- How do the results of this comparison vary across the other markets in the data set?